

Research and Practical Report

:theHarvester

Abstract

This document provides an academic and practical evaluation of theHarvester , an Open Source Intelligence(OSINT) framework designed for passive reconnaissance and foot printing. The report covers theoretical architecture, a functional breakdown of command-line interface (CLI) options, step-by-step verification procedures, and risk mitigation strategies.

Introduction

Within the first stages of ethical hacking or threat assessment, security personnel collect intelligence while keeping out of contact with the infrastructure of the target network whenever possible. theHarvester performs this process for you by extracting intelligence from publicly available sources such as search engines, domain name servers, data breaches, and threat intelligence APIs.

Key Capabilities:

- **Metadata Scrape (Passive):** Collects subdomains, IP addresses, virtual hosts, and employee email addresses.
- **API Call:** Calls threat intelligence services, such as Shodan, Censys, or VirusTotal.
- **DNS Enumeration (Active):** Enumerates subdomains through brute-forcing.

Command-Line Interface (CLI) Reference

The following table provides a reference for all available execution flags:

Flag	Argument	Description
-h, --help	N/A	Displays the standard help message and exits.
-d, --domain	DOMAIN	Target domain or company name (e.g., example.com).
-l, --limit	LIMIT	Maximum results to request per search engine (Default: 500).
-s, --start	START	Index of starting offset for pagination (Default: 0).
-p, --proxies	N/A	Enables proxy routing defined in proxies.yaml.
-s, --shodan	N/A	Queries Shodan for open ports on discovered hosts.
--screenshot	DIRECTORY	Saves web browser screenshots of active domains to a folder.
-e, --dns-server	DNS_SERVER	Custom DNS server IP for lookups.
-t, --take-over	N/A	Checks subdomains for dangling CNAME record takeovers.
-r, --dns-resolve	[RESOLVER]	Enables DNS resolution on discovered subdomains.
-n, --dns-lookup	N/A	Performs reverse DNS lookup on IP ranges.
-c, --dns-brute	N/A	Runs dictionary-based DNS brute-forcing.
-f, --filename	FILENAME	Saves output to <FILENAME>.xml and <FILENAME>.json.
-w, --wordlist	WORDLIST	Specifies a custom dictionary file for API endpoint scans.
-a, --api-scan	N/A	Scans discovered target hosts for active API endpoints.
-q, --quiet	N/A	Suppresses missing API key warnings.
-b, --source	SOURCE	Search engines/sources to query (or all).

Practical Testing and Validation Logs

Test 1. Basic Passive Search and Help Verification

Goal: Verify installation and perform a simple passive scan across publicsources.

Command:

- theHarvester -d rupeshkhadka1.com.np -l 400 -b all

```
(root@kali)-[~]
# theHarvester -d rupeshkhadka1.com.np -l 400 -b all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* theHarvester
*
* theHarvester 4.11.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: rupeshkhadka1.com.np

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverrun search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'rupeshkhadka1.com.np'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
```

[*] Searching Yahoo.
[*] Searching Waybackarchive.
[*] ASNS found: 1
AS13335
[*] Interesting Urls found: 1
https://rupeshkhadka1.com.np/
[*] No LinkedIn users found.
[*] LinkedIn Links found: 0
[*] IPs found: 10
104.21.87.170
172.64.80.1
172.67.144.197
2606:4700:3033::6815:57aa
2606:4700:3037::ac43:90c5
[*] No emails found.
[*] No people found.
[*] Hosts found: 1
*.rupeshkhadka1.com.np
[*] Performing SecurityScorecard scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.
[*] Performing BuiltWith scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for BuiltWith.

Test 2. Output File Generation (-f) and Quiet Mode(-q)

Goal: Verify that scan results export correctly into structured formats while hiding API key warnings.

Command:

- theHarvester -d rupeshkhadka1.com.np -l 200 -b all -q -f report_test

```
(root@kali)-[~]
# theHarvester -d rupeshkhadka1.com.np -l 200 -b all -q -f report_test
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* [theHarvester]
*
* theHarvester 4.11.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****
[*] Target: rupeshkhadka1.com.np

Failed to process bevigil search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for bevigil.

[!] Missing API key for Bitbucket.
Failed to process bufferoverrun search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for bufferoverrun.
Failed to perform BuiltWith search for word: 'rupeshkhadka1.com.np'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Failed to process brave search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for Brave Search.
Error initializing SearchGithubCode:
[!] Missing API key for Github.
[*] Mojeek: No API key found, using default scraping mode.

[!] Missing API key for SecurityScorecard.
[*] Searching Baidu.
Chaos API error: get access token here https://cloud.projectdiscovery.io
[*] Searching Chaos.
    Searching results.
[*] Searching Certspotter.
```

Verification Command:

- ls -l report_test.json report_test.xml

```
(root@kali)-[~]
# ls -l report_test.json report_test.xml
-rw-r--r-- 1 root root 276031 Jul 21 21:49 report_test.json
-rw-r--r-- 1 root root 397072 Jul 21 21:49 report_test.xml
```

```
(root@kali)-[~]  
# jq . report_test.json
```

```
{  
  "asns": [  
    "AS13335"  
  ],  
  "cmd": "-d rupeshkhadka1.com.np -l 200 -b all -q -f report_test",  
  "hosts": [  
    "*.rupeshkhadka1.com.np"  
  ],  
  "interesting_urls": [  
    "https://rupeshkhadka1.com.np/"  
  ],  
  "ips": [  
    "104.21.87.170",  
    "172.64.80.1",  
    "172.67.144.197",  
    "2606:4700:3033::6815:57aa",  
    "2606:4700:3037::ac43:90c5"  
  ],  
  "shodan": []  
}
```

```
(root@kali)-[~]  
# xmllint --format report_test.xml  
<?xml version="1.0" encoding="UTF-8"?>  
<theHarvester>  
  <cmd>-d rupeshkhadka1.com.np -l 200 -b all -q -f report_test</cmd>  
  <host>*.rupeshkhadka1.com.np</host>  
  <host>www.rupeshkhadka1.com.np</host>  
</theHarvester>
```

Ready For Delivery

- Order has been reached at delivery h
Sep 12, 2024 1:41 PM

- Order has been assigned to (Md. Ro
delivery

Sep 12, 2024 9:16 AM

Delivered

- Order has been delivered to the cust
Sep 13, 2024 2:12 PM

See time

Receipt

USE CASH ON DELIVERY

Product price

Test 3. Active DNS Resolution (-r) and Custom DNS server (-e)

Goal: Verify active DNS lookups for discovered subdomains using Cloudflare's public resolver(1.1.1.1).

Command:

- theHarvester -d rupeshkhadka1.com.np -b all -r -e 1.1.1.1

```
(root@kali)~# theHarvester -d rupeshkhadka1.com.np -b all -r -e 1.1.1.1
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*                               *
* [TheHarvester]                *
*                               *
* theHarvester 4.11.1           *
* Coded by Christian Martorella  *
* Edge-Security Research         *
* cmartorella@edge-security.com  *
*                               *
*****

[*] Target: rupeshkhadka1.com.np

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverrun search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'rupeshkhadka1.com.np'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
```

```
[*] ASNS found: 1
AS13335

[*] Interesting Urls found: 1
https://rupeshkhadka1.com.np/

[*] No LinkedIn users found.

[*] LinkedIn Links found: 0

[*] IPs found: 12
104.21.87.170
172.64.80.1
172.67.144.197
2606:4700:3033::6815:57aa
2606:4700:3037::ac43:90c5

[*] No emails found.
[*] No people found.
[*] Hosts found: 2
*.rupeshkhadka1.com.np:
www.rupeshkhadka1.com.np:104.21.87.170,172.67.144.197

[*] Performing SecurityScorecard scan ...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.

[*] Performing BuiltWith scan ...
```


Test 4. Subdomain Brute Forcing (-c) and Takeover Analysis(-t)

Goal: Validate dictionary-based subdomain enumeration and dangling evaluation.

Command:

- theHarvester -d rupeshkhadka1.com.np -b all -c -t

```
(root@kali)~# theHarvester -d rupeshkhadka1.com.np -b all -c -t
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* theHarvester 4.11.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: rupeshkhadka1.com.np

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverrun search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'rupeshkhadka1.com.np'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for Brave Search.

[+] ASNS found: 1
AS13335

[+] Interesting Urls found: 1
https://rupeshkhadka1.com.np/

[+] No LinkedIn users found.

[+] LinkedIn Links found: 0

[+] IPs found: 10
104.21.87.170
172.64.80.1
172.67.144.197
2606:4700:3033::6815:57aa
2606:4700:3037::ac43:90c5

[+] No emails found.
[+] No people found.
[+] Hosts found: 1
*.rupeshkhadka1.com.np

[+] Starting DNS brute force.
[Errno 2] No such file or directory: '/usr/lib/python3/dist-packages/theHarvester/data/wordlists/dns-names.txt'
```


Test 5. Automated Domain Screenshoting (--screenshot)

Goal: Capture visual evidence of HTTP/HTTPS web endpoints on resolved targets.

Command:

- theHarvester -d rupeshkhadka1.com.np -b all -r --screenshot ./screenshots_output

```
(root@kali)-[~]
└─$ theHarvester -d rupeshkhadka1.com.np -b all -r --screenshot ./screenshots_output
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* [TheHarvester]
*
* theHarvester 4.11.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****
[*] Target: rupeshkhadka1.com.np
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverrun search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'rupeshkhadka1.com.np'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'rupeshkhadka1.com.np'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

File "uvloop/loop.pyx", line 2841, in subprocess_exec
File "uvloop/loop.pyx", line 2799, in __subprocess_run
File "uvloop/handles/process.pyx", line 611, in uvloop.loop.UVProcessTransport.new
File "uvloop/handles/process.pyx", line 112, in uvloop.loop.UVProcess._init
FileNotFoundError: [Errno 2] No such file or directory
An exception has occurred while attempting to verify installation: [Errno 2] No such file or directory

Screenshots can be found in: ./screenshots_output/
Filtering domains for ones we can reach
Finished taking screenshots in 00:00 seconds
[+] Note there may be leftover chrome processes you may have to kill manually

[*] Performing SecurityScorecard scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.

[*] Performing BuiltWith scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for BuiltWith.

(root@kali)-[~]
└─$ ls -l screenshots_output
total 0
```

Test 6. API Scanning (-a) with Custom Wordlist (w)

Goal: Identify publicly exposed API endpoints across the discovered host using a wordlist.

Command: theHarvester -d rupeshkhadka1.com.np -b crtsh -a -w /usr/share/wordlists/dirb/common.txt

```
(root@kali)~# theHarvester -d rupeshkhadka1.com.np -b crtsh -a -w /usr/share/wordlists/dirb/common.txt
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.11.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****
[+] Target: rupeshkhadka1.com.np
[+] Searching CRTsh.
[+] No IPs found.
[+] No emails found.
[+] No people found.
[+] No hosts found.

[+] Starting API endpoint scanning with wordlist: /usr/share/wordlists/dirb/common.txt
2026-07-21 22:56:13,300 - api_endpoints - INFO - Starting API endpoint scan for rupeshkhadka1.com.np
2026-07-21 22:56:18,637 - api_endpoints - INFO - Detected schema for rupeshkhadka1.com.np: https
2026-07-21 22:56:18,637 - api_endpoints - INFO - Prepared 9250 endpoints to scan with concurrency 20
2026-07-21 22:56:23,971 - api_endpoints - WARNING - No status code received from response for URL: https://rupeshkhadka1.com.np/diary/
2026-07-21 22:56:23,990 - api_endpoints - WARNING - No status code received from response for URL: https://rupeshkhadka1.com.np/gif/
2026-07-21 22:56:24,013 - api_endpoints - WARNING - No status code received from response for URL: https://rupeshkhadka1.com.np/listusers/
2026-07-21 22:56:24,028 - api_endpoints - WARNING - No status code received from response for URL: https://rupeshkhadka1.com.np/alert/
2026-07-21 22:56:24,035 - api_endpoints - WARNING - No status code received from response for URL: https://rupeshkhadka1.com.np/resolve/
```

Defensive Remediation and Risk Analysis

TheHarvester exemplifies the capability of attackers to collect critical data from public sources. In order to reduce the amount of exposed data, it is advised to frequently audit DNS data, delete unused subdomains, protect employees' emails, secure API access, and employ security providers, such as Cloudflare and WAF. Regular monitoring of public assets will help mitigate potential risks associated with reconnaissance.

Conclusion

TheHarvester is a reliable OSINT tool for passive reconnaissance which can assist in collecting publicly available information on hosts, IP addresses, and domains. It was proven in practice that the tool can be used for intelligence gathering and report generation, thus becoming useful for penetration testing, security assessments, and defensive security applications.